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Fixed-Orientation Sliding Blocks

Contact Geometry, Guide Wrench, and the Common Boundary

A sliding centrifugal weight can move outward while two surfaces determine its axial position. The fixed cup carries one surface; the movable sheave carries the other. Once a guide prevents relative rotation of the weight, those two contact constraints can determine its translation from sheave closure alone. The resulting motion supplies the same effective mass and shaft inertia as any other mechanism admitted by the common boundary [1].

That reduction does not make the block torque-free. Contact forces generally have different moment arms about its centre of mass (COM), and the block still rotates in space with the shaft. This note derives the contact motion first, then the inertia functions, complete ideal-guide demand, and the impulses needed to retain the contacts at a velocity reset.

The supported model is a CVTech-style sliding-block arrangement. The examples are synthetic profiles, not measured CVTech hardware. A particular cup, weight, insert, moving sheave, and anti-rocking guide still require their actual geometry and mass properties before this model can represent that assembly.

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1 Attachments, coordinates, and retained motion

Use the right-handed local basis $(\boldsymbol{e}_r, \boldsymbol{e}_t, \boldsymbol{e}_z)$, with $\boldsymbol{e}_z$ along the shaft and $\boldsymbol{e}_t = \boldsymbol{e}_z \times \boldsymbol{e}_r$. The local sheave coordinate x is positive in closure; the moving contact track translates by $+x\boldsymbol{e}_z$. The lower track is axially fixed in the shaft frame. Both tracks and the orientation guide co-rotate with the shaft at angular speed ω and acceleration α.

Let $(R(x), Z(x))$ locate a datum fixed in the block. The block does not rotate relative to these local axes. Its COM offset from the datum is the fixed vector (c_r, c_t, c_z), so the COM coordinates are

$$r = R + c_r, \qquad y = c_t, \qquad z = Z + c_z. \tag{1.1}$$

Separating this datum from the COM is essential for tuning inserts: changing an insert can move the COM without moving either contacting face.

The complete identical set has mass M and contains n blocks. One block has mass M/n. Store the centroidal inertia tensor per unit mass as $\boldsymbol{K}$, expressed in (r, t, z) axes. Thus $\boldsymbol{I}_{C,\mathrm{set}} = M\boldsymbol{K}$ is the sum of corresponding local centroidal tensors, and $\boldsymbol{I}_{C,1} = M\boldsymbol{K}/n$ belongs to one block. Off-diagonal entries are the entries of the inertia tensor itself: for example, $K_{rz} = -\int u_r u_z \,\mathrm{d}m/m$. They are not unsigned geometric products of inertia.

The mass set is constant. The cup and rigid sheave masses remain in the host pulley accounting; they are not included in M. Springs and belt loads are separate elements. The contacts here are frictionless and unilateral. The initial guide is an explicit ideal constraint: it supplies a tangential force through the COM and an arbitrary holding couple, with no frictional loss. Physical guide capacities are unknown, rather than inferred from the outline.

2 Two finite profiles determine the body translation

Each contact face is a finite graph in a meridional section. Write the lower and upper block faces as $z = b_\ell(u)$ and $z = b_u(u)$, where u is radial position relative to the body datum. The two faces share a declared interval $[u_a, u_b]$, and $b_u > b_\ell$ throughout it. The fixed and moving tracks are $z = f_\ell(\rho)$ and $z = f_u(\rho) + x$, respectively, on finite radial domains. The whole block footprint must remain within both modelled track domains.

At a trial datum pose (R, Z), define the signed axial gaps

$$g_\ell(u; R, Z) = Z + b_\ell(u) - f_\ell(R + u), \tag{2.1}$$

$$g_u(u; R, Z, x) = f_u(R + u) + x - Z - b_u(u). \tag{2.2}$$

A nonnegative gap means separation without penetration. A retained contact requires the minimum gap on that face to be zero. This compares the full finite face, rather than selecting a tangent while ignoring another part of the same face that might penetrate the track.

The implementation supports lines, quadratics, and circular-arc graphs:

$$f(\rho) = f_0 + s(\rho - \rho_0) \qquad \text{(line)}, \tag{2.3}$$

$$f(\rho) = f_0 + s(\rho - \rho_0) + \tfrac{1}{2}k(\rho - \rho_0)^2 \qquad \text{(quadratic)}, \tag{2.4}$$

$$f(\rho) = f_0 + \sigma\big[a - \sqrt{a^2 - (\rho - \rho_0)^2}\big] \qquad \text{(circular arc)}, \tag{2.5}$$

with $a > 0$, $\sigma = \pm 1$, and the circle's domain strictly inside $|\rho - \rho_0| < a$. These are radial-coordinate graphs, not arbitrary rotated arcs or a general CAD representation. Overhangs and saddle contacts require a different contact construction.

2.1 Selecting a branch without hiding another minimum

The radial window $[R_{\min}, R_{\max}]$ is part of the input. Exact bounds on profile second derivatives determine whether each gap is convex or concave over this window. A convex gap has its minimum at an endpoint or at its unique stationary point; a concave gap has its minimum at an endpoint. An ambiguous curvature bound is rejected. Strict derivative or endpoint-gap inequalities then establish that the same endpoint or interior contact remains active throughout the window. Conformal contacts and mode switches are rejected; their pressure distribution or new branch cannot be obtained by blending the old motion curves.

To reduce the two equations, put

$$A(R) = \max_u [f_\ell(R + u) - b_\ell(u)], \qquad C(R) = \min_u [f_u(R + u) - b_u(u)]. \tag{2.6}$$

The lower contact gives $Z = A(R)$ and the upper contact gives

$$x = A(R) - C(R). \tag{2.7}$$

Disjoint track-slope bounds establish a nonzero derivative of this function throughout the selected window. A bracketed scalar solve then gives the unique R in that window. This is a statement about the declared branch window, not about every possible assembly elsewhere in the geometry.

Endpoint contact is allowed only when explicitly requested. It is an ideal point contact at a finite face endpoint with the smooth track's normal. A rounded nose of finite radius must be represented by its actual geometry if its change to the contact path matters. Neither an unknown corner radius nor a pressure centroid is fitted automatically.

3 Motion ratios from the same contact constraints

Let u_ℓ and u_u be the active face coordinates and let $s_i = f_i'(R + u_i)$ be the corresponding track slopes. At an interior tangent, $b_i'(u_i) = f_i'(R + u_i)$. The derivative of the minimum gap with respect to its minimizer vanishes; at a retained endpoint, u_i is constant. In either case, differentiating the two closed gaps gives

$$Z' = s_\ell R', \qquad Z' = 1 + s_u R'. \tag{3.1}$$

Consequently,

$$D = s_\ell - s_u, \qquad R' = \frac{1}{D}, \qquad Z' = \frac{s_\ell}{D}. \tag{3.2}$$

The constraint Jacobian has rows $(-s_\ell, 1)$ and $(s_u, -1)$. As D approaches zero the geometry ceases to determine a regular translation. The implementation bounds its condition number over the branch and rejects ill-conditioned windows rather than clipping the motion ratio.

Curvature changes the active track slope as the block moves. At an interior contact, differentiation of the tangency condition with respect to R yields

$$u_{i,R} = \frac{f_i''}{b_i'' - f_i''}, \qquad s_{i,R} = f_i''(1 + u_{i,R}). \tag{3.3}$$

At a retained endpoint $u_{i,R} = 0$. All profile derivatives in this expression are evaluated at the active material coordinates. Thus the second motion ratios are

$$R'' = -(s_{\ell,R} - s_{u,R})(R')^3, \qquad Z'' = s_\ell R'' + s_{\ell,R}(R')^2. \tag{3.4}$$

The fixed COM offset does not affect these derivatives: $r' = R'$, $z' = Z'$, and likewise for second derivatives.

4 Mass motion supplies the four quantities

Because orientation is fixed relative to the shaft frame, every material point has the same relative translation velocity $(r', 0, z')\dot{x}$. Its spin about the COM still contributes to shaft-rotation energy. Integrating the material velocity therefore gives

$$T_f = \tfrac{1}{2}M\left[(r')^2 + (z')^2\right]\dot{x}^2 - Mc_t r'\omega\dot{x} + \tfrac{1}{2}\left[M(r^2 + c_t^2) + MK_{zz}\right]\omega^2. \tag{4.1}$$

The mixed coefficient is $B_f = -Mc_t r'$. Fixed orientation alone does not remove it. The initial implementation requires $c_t = 0$; it does not infer cancellation from equally spaced blocks. With that condition established,

$$\boxed{\begin{aligned} M_f &= M[(r')^2 + (z')^2], \quad M_f' = 2M(r'r'' + z'z''), \\ J_f &= Mr^2 + MK_{zz}, \qquad J_f' = 2Mrr'. \end{aligned}} \tag{4.2}$$

These are total-set quantities. The count n does not multiply them again. The tensor must be finite and symmetric, with nonnegative principal inertias satisfying the triangle inequalities. These checks establish a possible inertia tensor; they do not establish that supplied mass properties actually belong to the specified outline and inserts.

The common CINDER element [1] consumes equation (4.2) and supplies

$$F_x = \tfrac{1}{2}J_f'\omega^2 - M_f\ddot{x} - \tfrac{1}{2}M_f'\dot{x}^2, \qquad \tau = -J_f\alpha - J_f'\dot{x}\omega. \tag{4.3}$$

There is one implementation of these relations in `CentrifugalInertiaForce`. The sliding-block provider adds the geometry and contact policy; it does not carry another transient force law. The same M_f and J_f enter kinetic energy and event projection.

5 Contact reactions and the torque on each block

The four maps describe work and kinetic energy, but their reduction has eliminated the individual constraint reactions. Recover those reactions from the block's free body before deciding whether the contact branch is admissible. The contact normals on the block, written as radial–axial pairs, are

$$\boldsymbol{n}_\ell = \frac{(-s_\ell, 1)}{\sqrt{1 + s_\ell^2}}, \qquad \boldsymbol{n}_u = \frac{(s_u, -1)}{\sqrt{1 + s_u^2}}. \tag{5.1}$$

With compression positive, the total corresponding normals satisfy

$$[\boldsymbol{n}_\ell\ \boldsymbol{n}_u]\begin{bmatrix} N_\ell \\ N_u \end{bmatrix} = M\begin{bmatrix} r'\ddot{x} + r''\dot{x}^2 - r\omega^2 \\ z'\ddot{x} + z''\dot{x}^2 \end{bmatrix}. \tag{5.2}$$

The tangential acceleration is separate. The ideal guide supplies

$$G_t = M(r\alpha + 2\omega r'\dot{x}). \tag{5.3}$$

The $2\omega r'\dot{x}$ term is the tangential acceleration associated with radial motion in the rotating frame. Neither contact normal in this model has a tangential component.

The guide must also prevent relative rocking. Let $\boldsymbol{a}_i$ run from the COM to contact i, let $\boldsymbol{N}_i = N_i(n_{i,r}, 0, n_{i,z})$, and use the full centroidal tensor. The required holding couple is

$$\boxed{\boldsymbol{G}_C = M\boldsymbol{K}\,(\alpha\boldsymbol{e}_z) + (\omega\boldsymbol{e}_z) \times [M\boldsymbol{K}(\omega\boldsymbol{e}_z)] - \sum_{i=\ell,u} \boldsymbol{a}_i \times \boldsymbol{N}_i.} \tag{5.4}$$

The spin terms have components

$$M(K_{rz}\alpha - K_{tz}\omega^2, \quad K_{tz}\alpha + K_{rz}\omega^2, \quad K_{zz}\alpha). \tag{5.5}$$

Even at constant shaft speed, a nonzero product of inertia can require a transverse couple. Even with zero products, the two contact moments need not cancel. A single shaft-axis gyration radius cannot determine this full demand.

For the straight synthetic case at $x = 7.5\,\text{mm}$, $r = 57\,\text{mm}$, $M = 0.75\,\text{kg}$ and $\omega = 350\,\text{rad}\,\text{s}^{-1}$, steady contact gives $F_x = 2094.75\,\text{N}$. The contact-only holding couple has magnitude $51.0595\,\text{N}\,\text{m}$ in corresponding local axes, or $17.0198\,\text{N}\,\text{m}$ per block for $n = 3$. Figure 1 shows why the lines of action do not remove this moment.

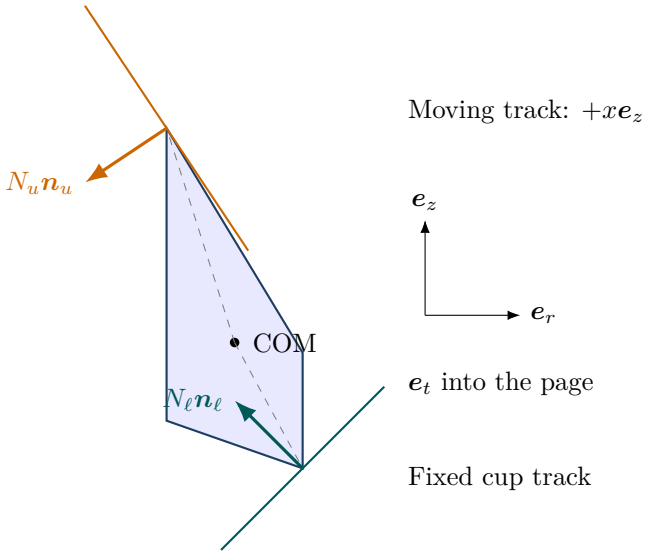


Figure 1: Local section of the straight synthetic block at the stated pose. The contact arms are measured from the COM. Shaft-induced centripetal acceleration is supplied by the resultant radial force; the guide couple must balance the contact moment as well as any centroidal spin demand. The track segments indicate contact surfaces, not complete cup or sheave solids.

All normals and guide components above are sums of corresponding local values; divide by n for one pocket. They are not a global transverse-vector sum around the pulley. Symmetric cancellation at assembly level does not unload each pocket. Circumferential sidewall forces alone also cannot resist a moment about the tangential axis: an anti-rocking feature must constrain that motion. A

specified real guide may transmit radial and axial forces too, in which case equation (5.2) must be replaced by the complete guide/contact equilibrium.

For the present ideal guide, the moving track delivers $F_x = -N_u n_{u,z}$. This agrees with equation (4.3) for the exact constraint motion. The runtime contact policy uses both N_ℓ and N_u at the actual solved accelerations. It rejects tension; it never replaces a negative normal by zero while retaining the same constrained mass map.

6 Retaining the contacts through a velocity reset

An instantaneous reset changes velocities at one fixed configuration. Smooth centrifugal, Coriolis, spring, and applied-torque terms do not supply finite impulses over that zero-duration interval. If the same two contacts and guide remain active, their required impulses follow from the velocity jumps:

$$[\boldsymbol{n}_\ell \ \boldsymbol{n}_u] \begin{bmatrix} P_\ell \\ P_u \end{bmatrix} = M \begin{bmatrix} r' \\ z' \end{bmatrix} \Delta\dot{x}, \tag{6.1}$$

$$P_t = Mr\Delta\omega, \tag{6.2}$$

$$\boldsymbol{L}_C = M\boldsymbol{K}(\Delta\omega \boldsymbol{e}_z) - \sum_i \boldsymbol{a}_i \times \boldsymbol{P}_i. \tag{6.3}$$

The normal impulses must be compressive. The ideal guide is permitted to carry its computed force impulse and couple impulse, but its real capacity is still unknown. A positive running normal force does not establish the sign of these impulses.

For the straight case, arresting $\dot{x} = +0.1\,\mathrm{m\,s^{-1}}$ with no shaft-speed jump requires

$$(P_\ell, P_u) = (-0.00848528,\ 0.04326662)\,\mathrm{N\,s}. \tag{6.4}$$

The lower contact would have to pull. The stop therefore cannot retain this branch, despite admissible pre-impact running forces. CINDER exposes the normal-impulse margins on candidate projections and rejects an inadmissible selected transition. Resolving what happens after release needs a different contact topology or an additional dynamic coordinate; this version does not invent that motion.

7 Compilation, inputs, and limits in CINDER 1.2.0

`SlidingBlockGeometry` stores the four profiles, radial branch window, local axial interval, and explicit endpoint-contact policy. `SlidingBlockMass` stores total mass, count, COM offset, and the full centroidal tensor per mass. `SlidingBlockSpec` combines these with the required `ideal_fixed_orientation` guide declaration. The saved component kind is `sliding_block_flyweight`. Distances are metres, mass is kilograms, profile curvature is inverse metres, and tensor entries are square metres. The three JSON examples adjacent to the package examples provide complete component records; they are not full CVT assemblies.

The compiler solves geometry outside the runtime RHS and constructs a common C^2 piecewise quintic Hermite representation of R, Z, u_ℓ, u_u. Values and first two derivatives come from the same contact construction. Both inertia functions and their derivatives use this one compiled motion.

Contact locations and normals also use the cached representation; no root solve is performed inside the closure iteration.

Starting from eight intervals, the compiler tests three interior points per cell and subdivides failed cells. Its absolute tolerances are $2 \times 10^{-10}\,\mathrm{m}$ for positions, 2×10^{-7} for first derivatives, and $2 \times 10^{-4}\,\mathrm{m}^{-1}$ for second derivatives. The compilation report records the largest observed errors. These are sampled interpolation diagnostics, distinct from the interval tests establishing the contact branch; they are not rigorous continuum bounds on interpolation error. The cache is keyed by the immutable geometry, so mass changes reuse the motion. Queries outside the compiled axial interval raise an error without extrapolation.

Limit	Implemented response
Invalid inputs	Reject non-finite values, impossible tensors, nonpositive thickness, invalid domains/count, and a nonzero tangential COM offset.
Geometric branch	Reject unsupported gap curvature, ambiguous/conformal contact, mode changes, insufficient track support, and ill-conditioned kinematics.
Operating state	Evaluate both normal forces with the solved local acceleration; reject contact loss. Report the complete ideal-guide demand separately.
Velocity reset	Resolve both normal impulses and reject a selected reset requiring tension. Direct projection callers must check the returned margins.
Hardware envelope	Speed, stress, pressure, real guide capacity, supporting-solid clearance, and three-dimensional interference require additional supplied data and checks; no universal safe RPM is derived from these profiles.

The mechanism may mount on either pulley when both contact tracks co-rotate with its shaft. A relative sheave helix is rejected. The host retains the full local-to-shared acceleration relation and the separate sheave/cup mass ownership. The frontend editor and authoritative preview service are a subsequent application change. The research viewer now imports CINDER's Python mechanics and contracts exported response coefficients in JavaScript; it does not maintain another contact-force equation.

8 Verification and what it establishes

The straight case recovers the force form used in Messick's straight-ramp model [4]: $M_f' = 0$ and the steady closing force is $Mrr'\omega^2$, with total M replacing a per-weight mass times count. This tests the force scale and geometry convention, not the dimensions of a particular machine.

The core smoke tests also differentiate inertial Cartesian trajectories of eight material masses. Only direct geometric positions are reused; the expected force, shaft torque and kinetic energy do not use the four-map formula. Three profile families, forward/reverse/zero shaft rotation, and both host shaft channels are exercised. Other checks cover derivative consistency, COM changes at fixed geometry, opposite products of inertia at identical four maps, saved-document/schema round trips, named running contacts, and valid and invalid stop projections. A short coupled neutral-shift transient checks primary angular-momentum conservation while axial motion changes shaft inertia.

The existing fixed-pivot launch regression remains a separate check of the common-boundary migration. None of these checks measures real CVTech guide capacity, contact pressure, wear,

or post-release motion. Their practical conclusion is that the stated constrained model supplies consistent continuous and impulsive mechanics and explicitly identifies conditions it cannot retain.

A Contact-coordinate derivatives for the interpolant

The compiler also needs derivatives of the material contact coordinate. With $h = u_R = f''/(b'' - f'')$ at an interior tangent, another differentiation gives

$$u_{RR} = \frac{f'''(1+h)^2 - b'''h^2}{b'' - f''}, \qquad u' = hR', \qquad u'' = hR'' + u_{RR}(R')^2. \tag{A.1}$$

All of these derivatives are zero for a retained material endpoint. Lines and quadratics have zero third derivative. For the circle used in section 2, with $d = \rho - \rho_0$,

$$f'' = \frac{\sigma a^2}{(a^2 - d^2)^{3/2}}, \qquad f''' = \frac{3\sigma a^2 d}{(a^2 - d^2)^{5/2}}. \tag{A.2}$$

The open-semicircle domain and branch checks keep these values finite. A contact-mode change requires a separate branch; this derivative construction does not assign a smoothed corner motion.

References

[1] *General Centrifugal Mechanisms: From Compatible Motion to Clamping Force and Shaft Coupling.* Revision 4; canonical CINDER 1.2.0 edition. Common-boundary document.

[2] *Clamping Mechanism Implementations: The Fixed-Pivot Roller Flyweight.* CINDER 1.2.0 edition. Fixed-pivot document.

[3] Kai Arseneau and W. Spencer Smith. *A First-Principles Dynamic Model of a Rubber V-Belt CVT for General Transient Simulation: With Mechanical Actuation and Traction-Limited Torque Transfer.* Supplied manuscript, October 1, 2026; Sections 3.2, 3.3 and 3.6.

[4] M. J. Messick. Supplied 2018 thesis, Section 9.4.1, printed page 109 (PDF page 118), Eq. 9.2. Used for the straight-ramp force limit; the dimensions in this note are synthetic.